

5-DOF 3D-Printed Robotic Arm (6-Axis Control)

Role: Mechatronics Engineering (Assembly, Power Management, C++ Programming)

Tech Stack: Arduino Mega 2560, OrcaSlicer, 3D Printing (PLA), C++

CHALLENGE AND OBJECTIVE

The goal of this project was to bridge the gap between software code and physical mechanics by building a functional robotic arm. It served as a hands-on introduction to 3D printing workflows, mechanical tolerances, and embedded control systems.

ARCHITECTURE AND HARDWARE

Electronic Components:

- **Microcontroller:** Arduino Mega 2560
- **Actuators:** 1× 28BYJ-48 (5V Stepper motor for precise base rotation) & 4x SG90 Micro Servomotors
- **Power Supply:** Custom-adapted 12A-5V external power supply
- **Control Interface:** 2x Analog Joysticks
- **Wiring:** 3x Terminal blocks (clemas)

Materials and Software:

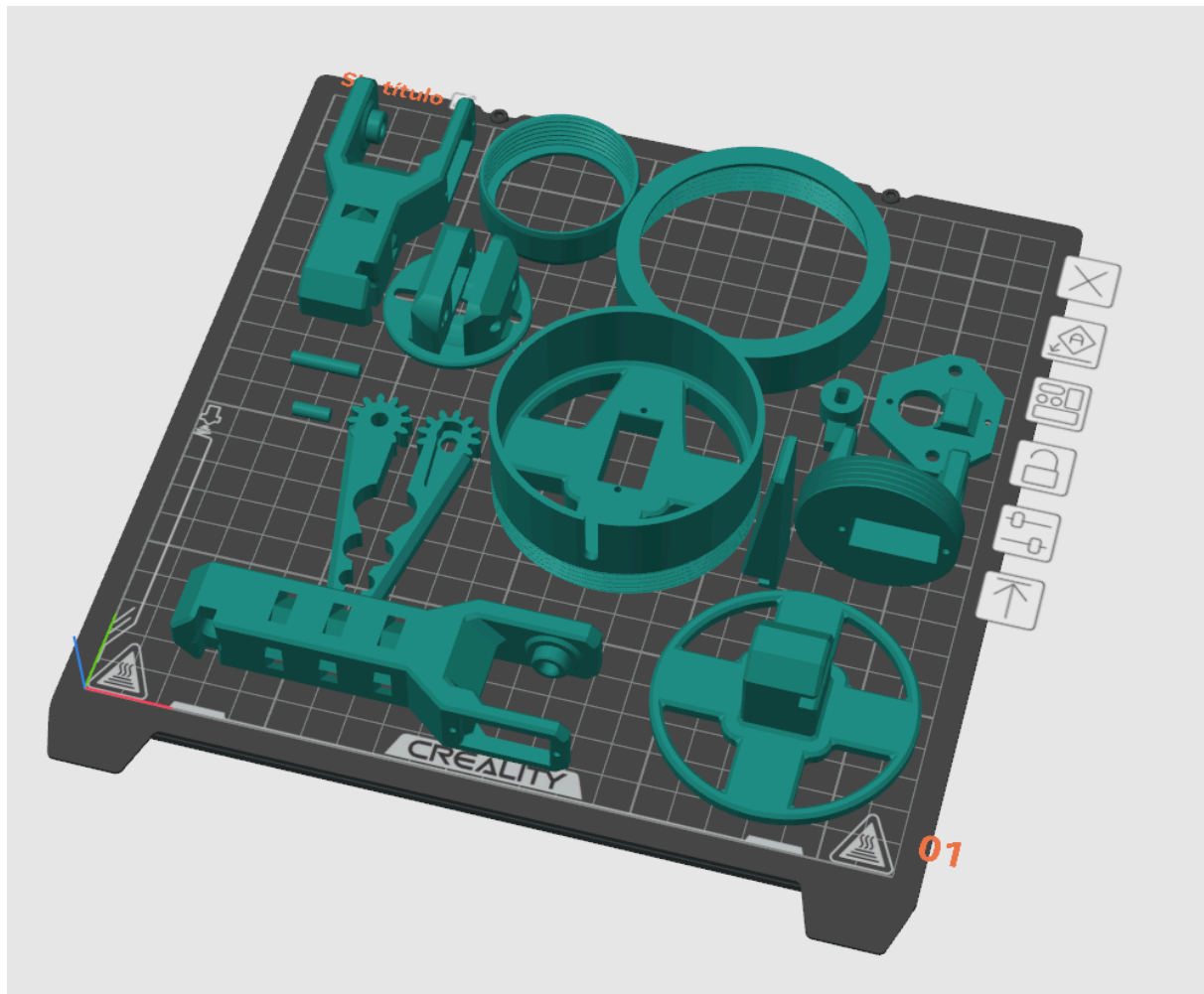
- **Software:** Arduino IDE, OrcaSlicer
 - **Manufacturing:** Black Basic PLA, standard mounting screws
 - **3D Model:** Sourced from [Printables.com](https://www.printables.com), adapted for the specific motors used.
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ENGINEERING PROCESS

1. 3D Printing and Manufacturing

The models were sliced using OrcaSlicer. During initial manufacturing, thermal warping became a significant issue.

- **Solution:** I optimized the printing workflow by dividing the components into smaller batches. This prevented rapid, uneven cooling between layers, completely eliminating the warping issue.



2. Circuit Design and Power Distribution

The most critical hardware challenge was powering the servomotors, which draw high peak amperage at 5V.

- **Solution:** Instead of buying a new module, I repurposed an unused 12A-5V barrel-jack power supply. I stripped the protective casing and adapted the raw wiring into terminal connector blocks. This provided a massive safety margin for the current draw while successfully recycling existing hardware.

3. Programming and Control

The control logic was written in C++ via the Arduino IDE. The system uses continuous angle mapping, incrementally adding or subtracting degrees from the servos' current position based on the joystick input.

TECHNICAL CHALLENGES & SOLUTIONS

- **Mechanical Tolerances (Thermal Deformation):**

Once printed, the servo cavities were too tight due to material expansion. Instead of reprinting, I applied targeted heat to the cavity edges to slightly soften the PLA. I then press-fit the servos, allowing the plastic to cool and harden around them for a perfectly rigid, custom fit

- **Noisy Analog Signals (Deadband Implementation):**

The analog joysticks fluctuated by ± 50 units on the 0-1023 analog scale, causing severe servo jitter. I implemented a software **deadband** (e.g., ignoring readings between 350 and 750) to filter out the baseline noise. This ensured the motors only moved upon deliberate joystick actuation.

- **Servo Gearbox Failure:**

During testing, internal gears inside one of the SG90 servos dislodged. I manually unscrewed the micro-servo casing, extracted the gear train, and carefully realigned each gear to its correct position, fully restoring the motor's functionality.

DEMONSTRATION AND RESULTS

The robotic arm successfully lifts a maximum payload of approximately 50 grams (in addition to its own structural weight). The movement is fluid, responsive, and operates with minimal vibration.

Future Roadmap:

- Integration of a camera module for computer vision and object recognition.
- Scaling up the project using high-torque metal-gear servos for heavy-duty applications.

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